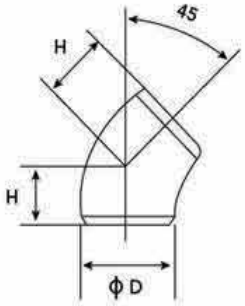
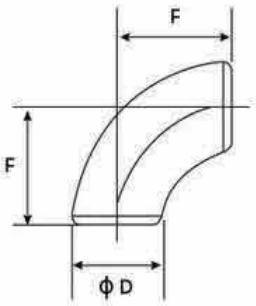
SGP / LG

45° & 90° ELBOW

JIS B2311 - 1997

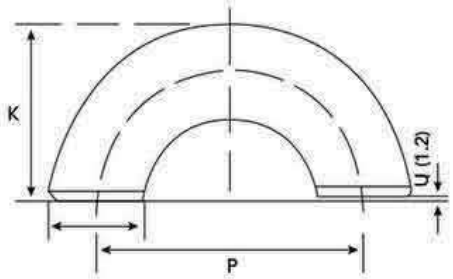


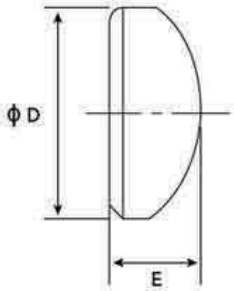


UNIT: mm

Nominal Diameter		Outside Diameter (D)	Dimension from the centerline to the end face			Approx. Weight (kg)		
			45° (H)	90° (F)		45°	90°	
A	B		Long	Long	Short	Long	Long	Short
15	½	21.7	15.8	38.1	-	0.039	0.078	-
20	¾	27.3	15.8	38.1	-	0.050	0.100	-
25	1	34.0	15.8	38.1	25.4	0.073	0.146	0.097
32	1 ¼	42.7	19.7	47.6	31.8	0.126	0.252	0.168
40	1 ½	48.6	23.7	57.2	38.1	0.175	0.350	0.233
50	2	60.5	31.6	76.2	50.8	0.317	0.634	0.423
65	2 ½	76.3	39.5	95.3	63.5	0.56	1.12	0.75
80	3	89.1	47.3	114.3	76.2	0.79	1.58	1.05
90	3 ½	101.6	55.3	133.4	88.9	1.06	2.11	1.41
100	4	114.3	63.1	152.4	101.6	1.46	2.91	1.94
125	5	139.8	78.9	190.5	127.0	2.25	4.49	2.99
150	6	165.2	94.7	228.6	152.4	3.55	7.09	4.73
200	8	216.3	126.3	304.8	203.2	7.20	14.40	9.60
250	10	267.4	157.8	381.0	254.0	12.70	25.40	16.93
300	12	318.5	189.4	457.2	304.8	19.05	38.10	25.40
350	14	355.6	220.9	533.4	355.6	28.35	56.70	37.80
400	16	406.4	252.5	609.6	406.4	37.15	74.30	49.53
450	18	457.2	284.1	685.8	457.2	47.10	94.20	62.79
500	20	508.0	315.6	762.0	508.0	59	117	78
550	22	558.8	347.2	838.2	558.8	71	141	94
600	24	609.6	378.7	914.4	609.6	84	168	112
650	26	660.4	410.3	990.6	660.4	99	198	132
700	28	711.2	441.9	1066.8	711.2	115	229	153
750	30	762.0	473.4	1143.0	762.0	132	264	176
800	32	812.8	505.0	1219.2	812.8	150	300	200
850	34	863.6	536.6	1295.4	863.6	169	338	225
900	36	914.4	568.1	1371.6	914.4	180	360	240
950	38	965.2	599.7	1447.8	965.2	212	424	283
1000	40	1016.0	631.2	1524.0	1016.0	235	470	313
1050	42	1066.8	662.8	1600.2	1066.8	259	518	345
1100	44	1117.6	694.4	1676.4	1117.6	284	569	379
1150	46	1168.4	725.9	1752.6	1168.4	311	622	415
1200	48	1219.2	757.5	1828.8	1219.2	339	677	451

SGP / LG
180° ELBOW & CAP
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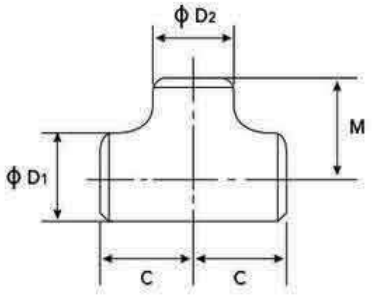




UNIT: mm

Nominal Diameter		Outside Diameter (D)	Centerline Dimension		Dimension from the back to the end face			Approx. Weight (kg)		
A	B		180° Elbow (P)		180° Elbow (K)		Caps (E)	180° Elbow (K)		Caps (E)
			Long	Short	Long	Short		Long	Short	
15	½	21.7	76.2	-	49.0	-	25.4	0.156	-	0.048
20	¾	27.2	76.2	-	51.7	-	25.4	0.200	-	0.058
25	1	34.0	76.2	50.8	55.1	42.4	38.1	0.292	0.195	0.098
32	1 ¼	42.7	95.2	63.6	69.0	53.2	38.1	0.504	0.336	0.139
40	1 ½	48.6	114.4	76.2	81.5	62.4	38.1	0.700	0.467	0.160
50	2	60.5	152.4	101.6	106.5	81.1	38.1	1.268	0.845	0.225
65	2 ½	76.3	190.6	127.0	133.5	101.7	38.1	2.24	1.49	0.300
80	3	89.1	228.6	152.4	158.9	120.8	50.8	3.16	2.11	0.490
90	3 ½	101.6	266.8	177.8	184.2	139.7	63.5	4.22	2.81	0.704
100	4	114.3	304.8	203.2	209.6	158.8	63.5	5.83	3.88	0.84
125	5	139.8	381.0	254.0	260.4	196.9	76.2	8.98	5.99	1.31
150	6	165.2	457.2	304.8	311.2	235.0	88.9	14.18	9.45	2.05
200	8	216.3	609.6	406.4	413.0	311.4	101.6	28.80	19.20	3.64
250	10	267.4	762.0	508.0	514.7	387.7	127.0	50.80	33.86	6.46
300	12	318.5	914.4	609.6	616.5	464.1	152.4	76.20	50.79	9.70
350	14	355.6	1066.8	711.2	711.2	533.4	165.1	113.40	75.59	13.26
400	16	406.4	1219.2	812.8	812.8	609.6	177.8	148.60	99.06	16.66
450	18	457.2	-	-	-	-	203.2	-	-	21.47
500	20	508.0	-	-	-	-	228.6	-	-	27.02

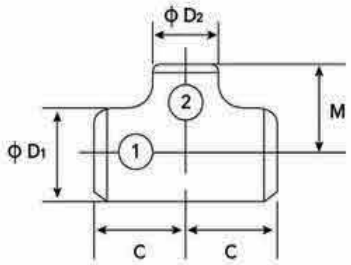
SGP / LG
STRAIGHT TEE
JIS B2311 - 1997

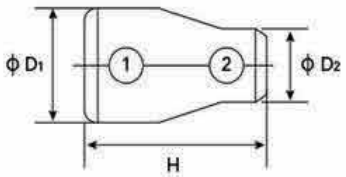


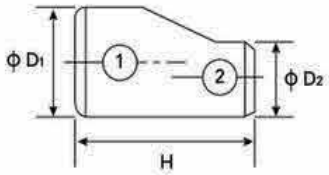
UNIT: mm

Nominal Diameter		Outside Diameter		Dimension from the centerline to the end face		Approx. Weight (kg)
A	B	D1	D2	C	C	
15	½	21.7	21.7	25.4	25.4	0.12
20	¾	27.2	27.2	28.6	28.6	0.17
25	1	34.0	34.0	38.1	38.1	0.32
32	1 ¼	42.7	42.7	47.6	47.6	0.56
40	1 ½	48.6	48.6	57.2	57.2	0.78
50	2	60.5	60.5	63.5	63.5	1.16
65	2 ½	76.3	76.3	76.2	76.2	1.94
80	3	89.1	89.1	85.7	85.7	2.55
90	3 ½	101.6	101.6	95.3	95.3	3.23
100	4	114.3	114.3	104.8	104.8	4.27
125	5	139.8	139.8	123.8	123.8	6.17
150	6	165.2	165.2	142.9	142.9	9.32
200	8	216.3	216.3	177.8	177.8	17.5
250	10	267.4	267.4	215.9	215.9	29.7
300	12	318.5	318.5	254.0	254.0	32.0
350	14	355.6	355.6	279.4	279.4	44.7
400	16	406.4	406.4	304.8	304.8	55.2
450	18	457.2	457.2	342.9	342.9	70.0
500	20	508.0	508.0	381.0	381.0	86.6
550	22	558.8	558.8	419.1	419.1	106
600	24	609.6	609.6	431.8	431.8	116
650	26	660.4	660.4	495.3	495.3	147
700	28	711.2	711.2	520.7	520.7	165
750	30	762.0	762.0	558.8	558.8	190
800	32	812.8	812.8	596.9	596.9	217
850	34	863.6	863.6	635.0	635.0	245
900	36	914.4	914.4	673.1	673.1	276
950	38	965.2	965.2	711.2	711.2	308
1000	40	1016.0	1016.0	749.3	749.3	342
1050	42	1066.8	1066.8	762.0	762.0	351
1100	44	1117.6	1117.6	812.8	812.8	395
1150	46	1168.4	1168.4	850.9	850.9	433
1200	48	1219.2	1219.2	889.0	889.0	474

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REDUCING TEE & REDUCER
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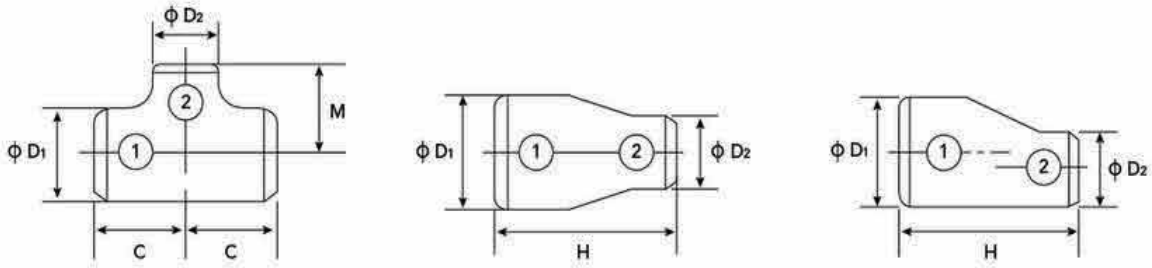




UNIT: mm

Nominal Diameter (1) X (2)		Outside Diameter		Overall Length	Dimension from the centerline to the end face		Approx. Weight (kg)	
A	B	D1	D2	H	C	M	Tee	Reducer
20 X 15	¾ X ½	27.2	21.7	38.1	28.6	28.6	0.16	0.057
25 X 20	1 X ¾	34.0	27.2	50.8	38.1	38.1	0.30	0.110
25 X 15	1 X ½	34.0	21.7	50.8	38.14	38.14	0.29	0.100
32 X 25	1 ¼ X 1	42.7	34.0	50.8	47.6	47.6	0.53	0.154
32 X 20	1 ¼ X ¾	42.7	27.2	50.8	47.6	47.6	0.50	0.140
32 X 15	1 ¼ X ½	42.7	21.7	50.8	47.6	47.6	0.49	0.130
40 X 32	1 ½ X 1 ¼	48.6	42.7	63.5	57.2	57.2	0.76	0.232
40 X 25	1 ½ X 1	48.6	34.0	63.5	57.2	57.2	0.72	0.209
40 X 20	1 ½ X ¾	48.6	27.2	63.5	57.2	57.2	0.68	0.191
40 X 15	1 ½ X ½	48.6	21.7	63.5	57.2	57.2	0.67	0.180
50 X 40	2 X 1 ½	60.5	48.6	76.2	63.5	60.3	1.08	0.364
50 X 32	2 X 1 ¼	60.5	42.7	76.2	63.5	57.2	1.04	0.344
50 X 25	2 X 1	60.5	34.0	76.2	63.5	50.8	0.99	0.315
50 X 20	2 X ¾	60.5	27.2	76.2	63.5	44.5	0.95	0.290
65 X 50	2 ½ X 2	76.3	60.5	88.9	76.2	69.9	1.78	0.594
65 X 40	2 ½ X 1 ½	76.3	48.6	88.9	76.2	66.7	1.70	0.543
65 X 32	2 ½ X 1 ¼	76.3	42.7	88.9	76.2	63.5	1.67	0.518
65 X 25	2 ½ X 1	76.3	34.0	88.9	76.2	57.2	1.61	0.480
80 X 65	3 X 2 ½	89.1	76.3	88.9	85.7	82.6	2.44	0.725
80 X 50	3 X 2	89.1	60.5	88.9	85.7	76.2	2.28	0.658
80 X 40	3 X 1 ½	89.1	48.6	88.9	85.7	73.0	2.21	0.611
80 X 32	3 X 1 ¼	89.1	42.7	88.9	85.7	69.9	2.17	0.590
90 X 80	3 ½ X 3	101.6	89.1	101.6	95.3	92.1	3.12	0.961
90 X 65	3 ½ X 2 ½	101.6	76.3	101.6	95.3	88.9	3.01	0.899
90 X 50	3 ½ X 2	101.6	60.5	101.6	95.3	82.6	2.85	0.826
90 X 40	3 ½ X 1 ½	101.6	48.6	101.6	95.3	79.4	2.77	0.771
90 X 32	3 ½ X 1 ¼	101.6	42.7	101.6	-	-	-	0.740
100 X 90	4 X 3 ½	114.3	101.6	101.6	104.8	101.6	4.09	1.17
100 X 80	4 X 3	114.3	89.1	101.6	104.8	98.4	3.98	1.10
100 X 65	4 X 2 ½	114.3	76.3	101.6	104.8	95.3	3.87	1.04
100 X 50	4 X 2	114.3	60.5	101.6	104.8	88.9	3.71	0.967
100 X 40	4 X 1 ½	114.3	48.6	101.6	104.8	85.7	3.63	0.910

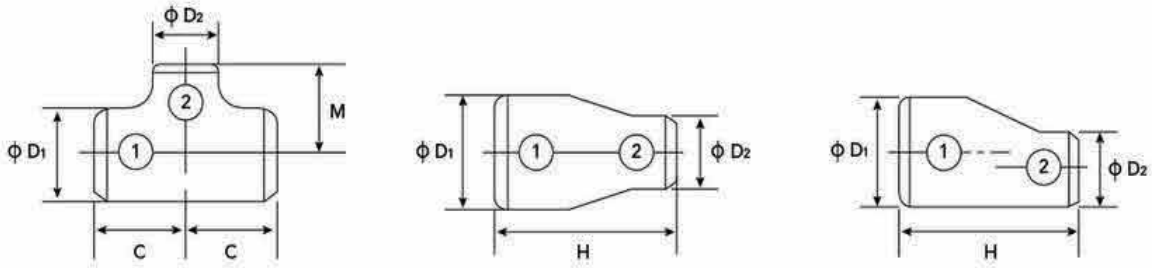
SGP / LG
REDUCING TEE & REDUCER
JIS B2311 - 1997



UNIT: mm

Nominal Diameter (1) X (2)		Outside Diameter		Overall Length	Dimension from the centerline to the end face		Approx. Weight (kg)	
A	B	D1	D2	H	C	M	Tee	Reducer
125 X 100	5 X 4	139.8	114.3	127.0	123.8	117.5	5.86	1.74
125 X 90	5 X 3 ½	139.8	101.6	127.0	123.8	114.3	5.68	1.65
125 X 80	5 X 3	139.8	89.1	127.0	123.8	111.1	5.56	1.58
125 X 65	5 X 2 ½	139.8	76.3	127.0	123.8	108.0	5.45	1.50
125 X 50	5 X 2	139.8	60.5	127.0	123.8	104.8	5.32	1.41
150 X 125	6 X 5	165.2	139.8	139.7	142.9	136.5	8.80	2.55
150 X 100	6 X 4	165.2	114.3	139.7	142.9	130.2	8.49	2.36
150 X 90	6 X 3 ½	165.2	101.6	139.7	142.9	127.0	8.31	2.27
150 X 80	6 X 3	165.2	89.1	139.7	142.9	123.8	8.19	2.18
150 X 65	6 X 2 ½	165.2	76.3	139.7	142.9	120.7	8.08	2.09
200 X 150	8 X 6	216.3	165.2	152.4	177.8	168.3	16.2	4.17
200 X 125	8 X 5	216.3	139.8	152.4	177.8	161.9	15.7	3.87
200 X 100	8 X 4	216.3	114.3	152.4	177.8	155.6	15.4	3.67
200 X 90	8 X 3 ½	216.3	101.6	152.4	177.8	152.4	15.2	3.56
250 X 200	10 X 8	267.4	216.3	177.8	215.9	203.2	27.8	6.87
250 X 150	10 X 6	267.4	165.2	177.8	215.9	193.7	26.6	6.32
250 X 125	10 X 5	267.4	139.8	177.8	215.9	190.5	26.1	6.06
250 X 100	10 X 4	267.4	114.3	177.8	215.9	184.2	25.8	5.80
300 X 250	12 X 10	318.5	267.4	203.2	254.0	241.3	41.5	9.97
300 X 200	12 X 8	318.5	216.3	203.2	254.0	228.6	39.6	9.29
300 X 150	12 X 6	318.5	165.2	203.2	254.0	219.1	38.3	8.69
300 X 125	12 X 5	318.5	139.8	203.2	254.0	215.9	37.9	8.39
350 X 300	14 X 12	355.6	318.5	330.2	279.4	269.9	42.7	21.2
350 X 250	14 X 10	355.6	267.4	330.2	279.4	257.2	41.2	19.7
350 X 200	14 X 8	355.6	216.3	330.2	279.4	247.7	40.0	18.3
350 X 150	14 X 6	355.6	165.2	330.2	279.4	238.1	39.0	16.9
400 X 350	16 X 14	406.4	355.6	355.6	304.8	304.8	54.2	25.9
400 X 300	16 X 12	406.4	318.5	355.6	304.8	295.3	52.2	24.7
400 X 250	16 X 10	406.4	267.4	355.6	304.8	282.6	50.7	23.2
400 X 200	16 X 8	406.4	216.3	355.6	304.8	273.1	49.4	21.7
400 X 150	16 X 6	406.4	165.2	-	304.8	263.5	48.5	-

SGP / LG
REDUCING TEE & REDUCER
JIS B2311 - 1997

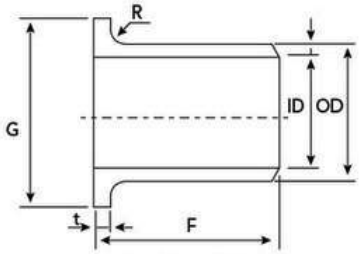


UNIT: mm

Nominal Diameter (1) X (2)		Outside Diameter		Overall Length	Dimension from the centerline to the end face		Approx. Weight (kg)	
A	B	D1	D2	H	C	M	Tee	Reducer
450 X 400	18 X 16	457.2	406.4	381.0	342.9	330.2	67.9	31.5
450 X 350	18 X 14	457.2	355.6	381.0	342.9	330.2	66.9	29.8
450 X 300	18 X 12	457.2	318.5	381.0	342.9	320.7	64.9	28.6
450 X 250	18 X 10	457.2	267.4	381.0	342.9	308.0	63.4	27.1
500 X 450	20 X 18	508.0	457.2	508.0	381.0	368.3	84.2	47.0
500 X 400	20 X 16	508.0	406.4	508.0	381.0	355.6	82.1	44.7
500 X 350	20 X 14	508.0	355.6	508.0	381.0	355.6	81.1	42.4
500 X 300	20 X 12	508.0	318.5	508.0	381.0	346.1	79.1	40.8
500 X 250	20 X 10	508.0	267.4	-	381.0	333.4	77.6	-
500 X 200	20 X 8	508.0	216.3	-	381.0	323.9	76.3	-
550 X 500	22 X 20	558.8	508.0	508.0	419.1	406.4	103	52.0
550 X 450	22 X 18	558.8	457.2	508.0	419.1	393.7	101	49.7
550 X 400	22 X 16	558.8	406.4	508.0	419.1	381.0	98.9	47.5
550 X 350	22 X 14	558.8	355.6	508.0	-	-	-	45.3
600 X 550	24 X 22	609.6	558.8	508.0	431.8	431.8	115	57.1
600 X 500	24 X 20	609.6	508.0	508.0	431.8	431.8	114	54.8
600 X 450	24 X18	609.6	457.2	508.0	431.8	419.1	111	52.6
600 X 400	24 X 16	609.6	406.4	508.0	-	-	-	50.4
650 X 600	26 X 24	660.4	609.6	609.6	495.3	482.6	144	74.5
650 X 550	26 X 22	660.4	558.8	609.6	495.3	469.9	141	71.7
650 X 500	26 X 20	660.4	508.0	609.6	495.3	457.2	138	68.9
650 X 450	26 X 18	660.4	457.2	609.6	-	-	-	66.3
700 X 650	28 X 26	711.2	660.4	609.6	520.7	520.7	164	80.5
700 X 600	28 X 24	711.2	609.6	609.6	520.7	508.0	161	77.7
700 X 550	28 X 22	711.2	558.8	609.6	520.7	495.3	158	75.0
700 X 500	28 X 20	711.2	508.0	609.6	-	-	-	72.4
750 X 700	30 X 28	762.0	711.2	609.6	558.8	546.1	188	86.6
750 X 650	30 X 26	762.0	660.4	609.6	558.8	546.1	185	83.8
750 X 600	30 X 24	762.0	609.6	609.6	558.8	533.4	182	81.1
750 X 550	30 X 22	762.0	558.8	609.6	-	-	-	78.5
800 X 750	32 X 30	812.8	762.0	609.6	596.9	584.2	213	92.6
800 X 700	32 X 28	812.8	711.2	609.6	596.9	571.5	210	89.8

STUB END

GB/T 12459, GB/T 13401
ASME / ASTM ANSI B16.9
MSS - SP43
TYPE A/B



Lap Joint Stub End

UNIT: mm

Nominal Diameter		Outside Diameter of Cylinder (OD)		Overall Length F		Radius Fillet R		Diameter of Lap
DN	NPS	Max	Min	Long Type	Short Type	A	B_{max}	G
15	½	22.8	20.5	76	51	3	0.8	35
20	¾	28.1	25.9	76	51	3	0.8	43
25	1	35.0	32.6	102	51	3	0.8	51
32	1 ¼	43.6	41.4	102	51	5	0.8	64
40	1 ½	49.9	47.5	102	51	6	0.8	73
50	2	62.4	59.5	152	64	8	0.8	92
65	2 ½	75.3	72.2	152	64	8	0.8	105
80	3	91.3	88.1	152	64	10	0.8	127
90	3 ½	104.0	100.8	152	76	10	0.8	140
100	4	116.7	113.5	152	76	11	0.8	157
125	5	144.3	140.5	203	76	11	1.6	186
150	6	171.3	167.5	203	89	13	1.6	216
200	8	222.1	218.3	203	102	13	1.6	270
250	10	277.2	272.3	254	127	13	1.6	324
300	12	328.0	323.1	254	152	13	1.6	381
350	14	359.9	354.8	305	152	13	1.6	413
400	16	411.0	405.6	305	152	13	1.6	470
450	18	462.0	456.0	305	152	13	1.6	533
500	20	514.0	507.0	305	152	13	1.6	584
550	22	565.0	558.0	305	152	13	1.6	641
600	24	616.0	609.0	305	152	13	1.6	692

Thickness refer to Page 6 ANSI PIPE SCHEDULE